

Bayes' Rule Applied

Video Lesson

Duration: 00:13:11

Slide Deck

M11S05 (<https://canvas.tamu.edu/courses/430127/files/92570691?wrap=1>) 

Lesson Transcript

In this lecture, we look at some applications of the Bayes' Rule.

Diagnostic vs. Causal Knowledge

Consider these probabilities:

- $P(\text{Symptom}|\text{Disease})$: **causal knowledge**
 - relatively fixed.
- $P(\text{Disease})$: somewhat variable.
- $P(\text{Disease}|\text{Symptom})$: **diagnostic knowledge**
 - fluctuates as $P(\text{Disease})$ change.

$P(\text{Disease}|\text{Symptom})$ directly measured can be no longer accurate when $P(\text{Disease})$ changes (e.g. an epidemic outburst), however the calculation based on Bayes' rule can be much more robust.

First, let us consider different types of knowledge and how Bayes' Rule can help.

In the context of probabilistic inference, we can consider two types of knowledge, diagnostic and causal knowledge.

First, let us consider causal knowledge. This can be expressed in this conditional probability. There is a probability of a symptom given the disease. Because the disease is the cause of the symptom, we are following this in the causal direction.

Due to this being causal, this conditional probability is relatively fixed.

Next, we have the prior probability of a disease. This may somewhat change over time. For example, it could be a seasonal disease, or maybe a cure could have been found.

Finally, we have diagnostic knowledge. This is expressed as the conditional probability. Probability of disease given symptom. Since we are looking at the symptoms and based on that, we are trying to infer the disease, which is diagnostic knowledge.

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- $P(\text{Disease}|\text{Symptom})$: **diagnostic knowledge**
- fluctuates as $P(\text{Disease})$ change.

$$P(\text{Disease}|\text{Symptom}) = \frac{P(\text{Symptom}|\text{Disease}) P(\text{Disease})}{P(\text{Symptom})}$$

$P(\text{Disease}|\text{Symptom})$ directly measured can be no longer accurate when $P(\text{Disease})$ changes (e.g. an epidemic outburst), however the calculation based on Bayes' rule can be much more robust.

This conditional probability, the probability of disease given symptom, is somehow variable because when we write this down using Bayes' Rule, we get this. The probability of disease given symptom equals the probability of symptom given disease multiplied by the probability of disease. And as we have seen in the second line, the probability of disease is variable.

So, as a whole, this whole thing is variable. And part of the reason is that this probability of symptom-given disease is fixed, and this is the only variable quantity. So, this may vary accordingly.

We can attempt to measure or estimate this probability, the probability of disease given symptom directly, by observing these cases, but it will no longer be accurate when the prior probability, probability of disease, changes.

So, we have to estimate this again. Whereas if we try to estimate this using the Bayes' Rule, then we have this fixed causal knowledge, and we only need to measure the probability of disease, the prior probability, to update this estimate quickly. So, using the calculation based on Bayes' Rule can be much more robust.

Comparison of Conditional Probabilities

When C_1 or C_2 can cause phenomenon (or *effect*) E , to find out the which is the more probable cause of phenomenon E , we do not need to explicitly calculate $P(E)$:

- $P(C_1|E) = \frac{P(E|C_1)P(C_1)}{P(E)}$
- $P(C_2|E) = \frac{P(E|C_2)P(C_2)}{P(E)}$
- From the above, we get:

$$\frac{P(C_1|E)}{P(C_2|E)} = \frac{P(E|C_1)P(C_1)}{P(E|C_2)P(C_2)} = \frac{a}{b}$$

As we can see, Bayes' Rule gives us this form where we have to figure out this total probability, that is, the quantity that goes in the denominator. But computing that can be challenging. But one way of using the Bayes' Rule is to eliminate the need to estimate that. And this, of course, when we have two causes, for example, and we are only interested in whether the first course, C_1 , or the second course, C_2 , is the more probable one.

In this case, let us say there is a common effect from which we want to determine whether the cause is C_1 , or the cause is C_2 . So, we can write down the conditional probability for the first one as the probability of Cause 1 given effect. And for the second one, the probability becomes the probability of Cause 2 given effect.

And again, using the Bayes' Rule, we get this for the first case, and we get this for the second case. Now, since we are interested in which one is more probable, we can take the ratio of these two. So, when we take the ratio of these two, then the denominator cancels out, and we have this ratio: the probability of effect given Cause 1 times the probability of Cause 1, divided by the probability of effect given Cause 2 multiplied by the probability of Cause 2.

Let us say this is a , and this is b . Then the ratio between a and b could either be equal to 1, greater than 1, or less than 1. And depending on the outcome, we can say C_1 or C_2 is more probable.

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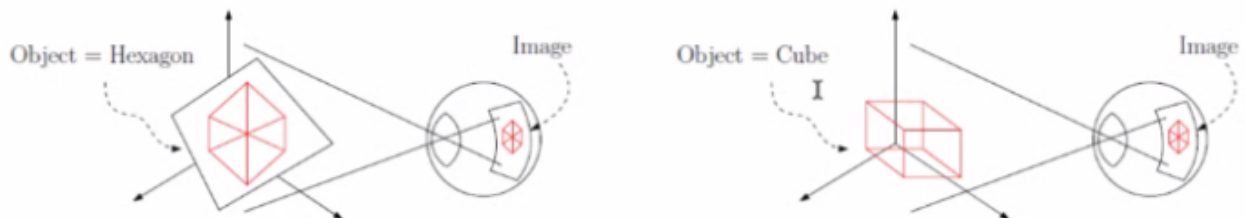
$$\frac{P(C_1|E)}{P(C_2|E)} = \frac{P(E|C_1)P(C_1)}{P(E|C_2)P(C_2)} = \frac{a}{b}$$

if $a/b > 1$,
 C_1 is the more probable cause

if $a/b < 1$
 C_2 is the more probable cause

To repeat, if a / b is greater than 1, C_1 is the more probable cause. And if a / b is less than 1, then C_2 is the more probable cause.

Example: The Problem of Object Recognition



Given an image projected on the retina, what is the more likely cause? the 2D hexagon? or a transparent 3D cube? This is basically a computer vision problem.

$$\frac{P(\text{Hexagon}|\text{Image})}{P(\text{Cube}|\text{Image})} = \frac{P(\text{Image}|\text{Hexagon})P(\text{Hexagon})}{P(\text{Image}|\text{Cube})P(\text{Cube})} = \frac{a}{b}$$

A probabilistic vision agent can make a decision based on such a ratio.

Let us apply that to the simple domain where we want to recognize what the object is given the projection of that object in the retina. So, here is the eyeball, and at the back of our eyeball is the retina with the photoreceptors.

So, we can present some object, in this case, a 2-D plane or sheet of paper with a hexagon image on it, and we will get this kind of hexagon shape projected on our retina. And based on that, we want to determine whether the object is a hexagon or something else.

On the other hand, when we have this transparent cube in a wireframe-like shape. If we look at it

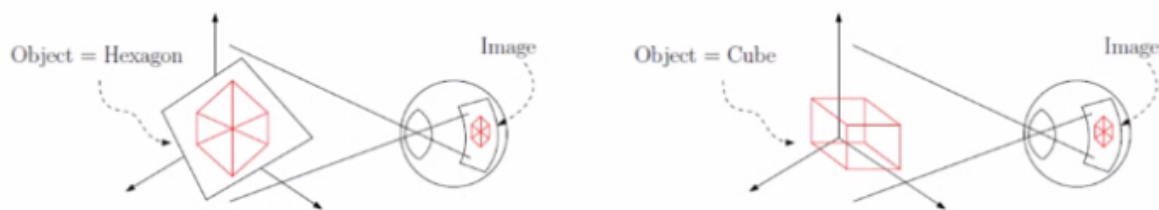
from a particular angle, then it may actually project something that looks like a hexagon onto the retina. So, when we just look at this projection, we are confused. Should it be a hexagon on a 2-D sheet of paper, or is it a 3-D cube? So, given an image projected onto the retina, we want to answer what is the more likely cause?

Is it the 2-D hexagon, or is it the transparent 3-D cube? This is basically a computer vision problem. Given an image, we want to know what is done underlying reality.

Using the same ratio idea, we find the ratio between the probability of the object being a hexagon given the image divided by the probability of the object being a cube given the image. It boils down to this after applying Bayes' Rule with the probability of the image given a hexagon times the probability of a hexagon divided by the probability image given a cube times the probability of a cube.

And let us say it comes out to a / b .

Example: Object Recognition (cont'd)



$$\frac{P(\text{Hexagon}|\text{Image})}{P(\text{Cube}|\text{Image})} = \frac{P(\text{Image}|\text{Hexagon})P(\text{Hexagon})}{P(\text{Image}|\text{Cube})P(\text{Cube})} = \frac{a}{b}$$

- Decision: if $a/b > 1$, it is most likely that a hexagon generated the image. If $a/b < 1$, it is most likely that a cube generated the image.

If, again, a / b is greater than 1, it is more likely to be the hexagon, the 2-D hexagon, that is the underlying object. If a / b is less than 1, it is most likely the cube that generated the retinal projection.

Of course, this ratio depends heavily on both of these, the likelihood and the prior. So, if in the world that we live in, we more normally see a lot of hexagons like that as opposed to cubes, then regardless of the likelihood, a would be much greater than b .

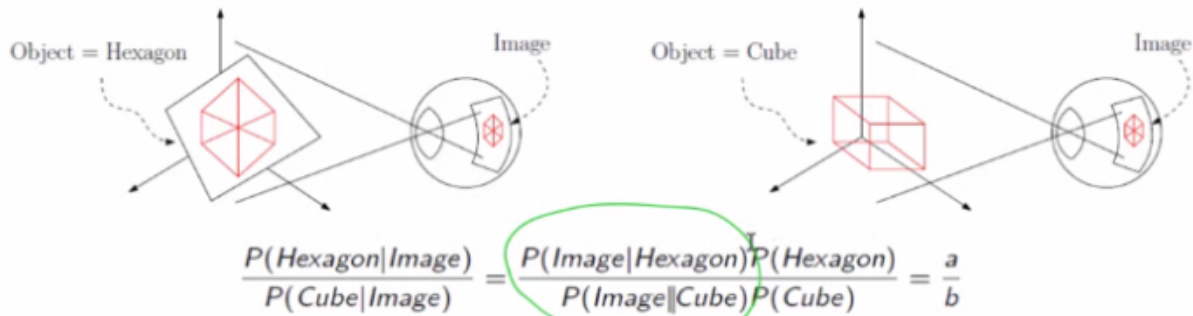
So, given that we say when we receive this information on our retina, the numerator part is greater than the denominator, our answer would be a hexagon, the 2-D hexagon. Whereas if we live in a world where we rarely see these, this prior probability for the cube is very high compared to the

prior probability of the hexagon.

Then again, b would be much greater than a , so our conclusion would be a . So, it is a cube.

Now consider the case where the P of a hexagon is the same as the P of a cube.

Example: Object Recognition (cont'd)



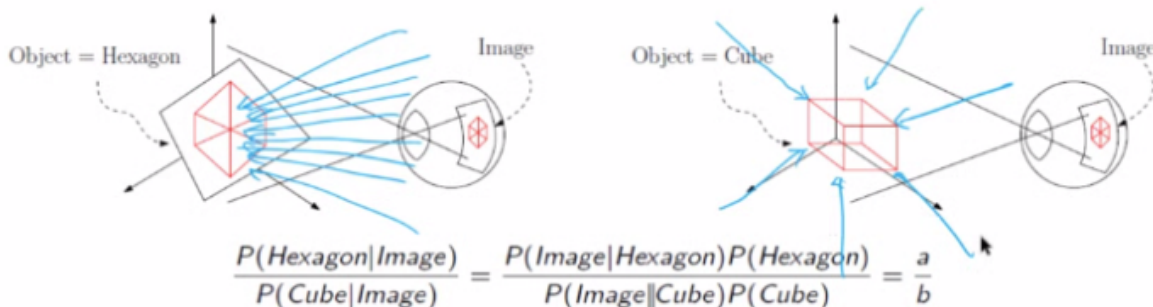
- Decision: if $a/b > 1$, it is most likely that a hexagon generated the image. If $a/b < 1$, it is most likely that a cube generated the image.

$$p(\text{Hexagon}) = p(\text{Cube})$$

So, the prior probabilities are the same.

In this case, we must consider the ratio between the two likelihood functions. That is, the probability image given hexagon is divided by the probability image given cube.

Example: Object Recognition (cont'd)



- Decision: if $a/b > 1$, it is most likely that a hexagon generated the image. If $a/b < 1$, it is most likely that a cube generated the image.

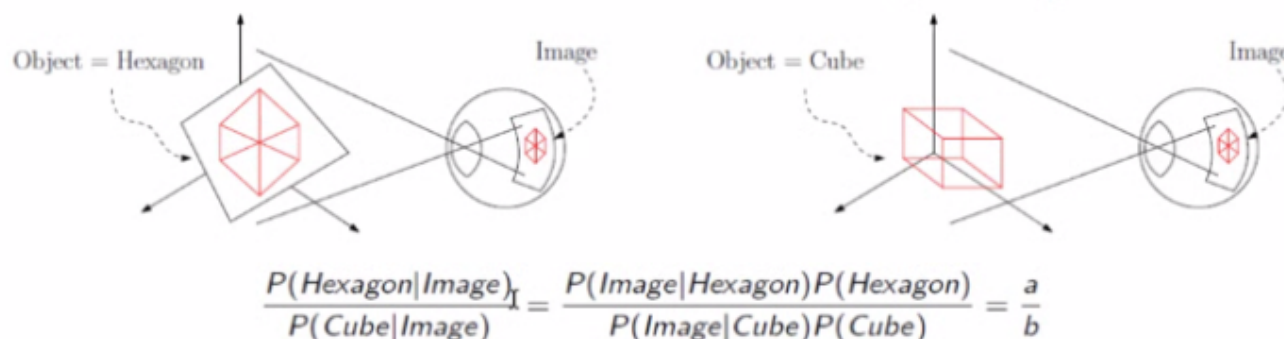
$$p(\text{Image}|\text{Hexagon}) \gg p(\text{Image}|\text{Cube})$$

We can consider this illustration when we have a hexagon on a 2-D plane like that, then viewing it from many different angles would result in this kind of image. Whereas for the cube, there are only eight different directions that we can look at that correspond to the eight corners of the cube, which will result in this kind of hexagon shape.

Consider this particular rendering. There is no center where all of these lines meet. And this would be the normal view from any other angle. So, we can just reason that the probability of an image, given a hexagon, with this particular image being this particular shape, is much greater than the probability of this kind of image, given a cube.

Hence, when we just consider this ratio, we can see that this is much greater than 1. As a result, we can conclude that when we have this image, the more probable cause is a hexagon.

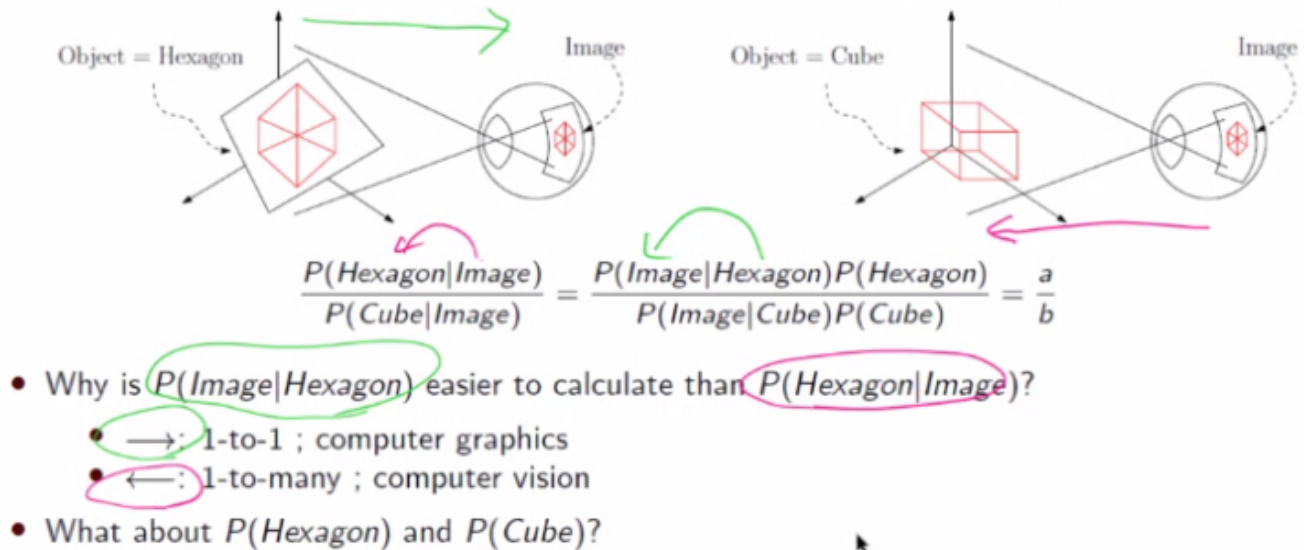
Example: Object Recognition (cont'd)



- Why is $P(\text{Image}|\text{Hexagon})$ easier to calculate than $P(\text{Hexagon}|\text{Image})$?
 - $\rightarrow$: 1-to-1 ; computer graphics
 - $\leftarrow$: 1-to-many ; computer vision
- What about $P(\text{Hexagon})$ and $P(\text{Cube})$?

Finally, let us consider this question, why is the probability of an image given a hexagon easier to calculate than the probability of a hexagon given an image? The whole reason why we turn this probability of a hexagon-given image into the product of the likelihood, the probability of an image given hexagon times the prior, is because this conditional probability is easier to estimate than that one.

Example: Object Recognition (cont'd)



Let us consider this first one, the probability of an image given a hexagon. We are given a hexagon, and we want to determine what the image is. That corresponds to this forward direction. Given this image of a hexagon, we want to find out what is the projected image.

Whereas given the projected image, trying to determine the hexagon object is actually going backward.

We note that for the forward direction, given an object, there is only one unique projection, so the mapping is 1 to 1. Whereas for the backward direction, given the projected image, there could be multiple different causes. Just using these examples, it is 1 to 2, but in fact, if we arrange these lines very carefully, then viewing them from a very specific angle can result in this kind of projected image.

It is, in fact, 1 to many or even 1 to infinity.

This first forward direction is computer graphics. Given an object in 3-D, we want to render it. And this backward direction is that of computer vision. Given an image, we want to identify what is the underlying object.

Example: Object Recognition (cont'd)



$$\frac{P(\text{Hexagon}|\text{Image})}{P(\text{Cube}|\text{Image})} = \frac{P(\text{Image}|\text{Hexagon})P(\text{Hexagon})}{P(\text{Image}|\text{Cube})P(\text{Cube})} = \frac{a}{b}$$

- Why is $P(\text{Image}|\text{Hexagon})$ easier to calculate than $P(\text{Hexagon}|\text{Image})$?
 - $\rightarrow$: 1-to-1 ; computer graphics
 - $\leftarrow$: 1-to-many ; computer vision
- What about $P(\text{Hexagon})$ and $P(\text{Cube})$?

Due to this kind of ambiguity, this backward direction is much more difficult to compute. As we mentioned, we only need to take care of the prior probabilities, the probability of a hexagon or the probability of a cube, and this will somehow get estimated as we experience the world.

In the following lecture, we will look at Bayesian updating.